Mid Sweden University (Mittuniversitetet)

Omnisound

User-Centred Design and Evaluation
of a Unified Audio Interface

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Human-Computer Interaction / Interaction Design

Project group: Group 1

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Abstract

People often listen to music, podcasts and audiobooks, but keeping track of saved and unfinished content across audio applications can be inconvenient. This project explored how these different types of audio could be organised in one interface without making navigation confusing. We developed Omnisound, a desktop-first web prototype with shared search, a Library for saved items and a Continue Listening section for unfinished audio.

We began with a project concept and paper sketches, then built an interactive first prototype (V1). We tested it with three students using three tasks: finding and saving Weezer, resuming an unfinished audiobook and browsing podcasts. We recorded task completion, time, navigation routes and participant comments. Based on the observations, we created a second prototype (V2) with a smaller Home banner, a more noticeable Continue Listening section and clearer Save/Saved controls. Three new students tested V2 using the same tasks.

All nine task attempts were completed in each round without direct help. However, only one of three V1 participants used Continue Listening directly, compared with all three V2 participants. The average time for resuming the audiobook was 27.0 seconds in V1 and 13.3 seconds in V2. The other tasks also had shorter recorded times in the second round.

The results suggest that the changes made unfinished audio easier to find for these participants. Because the tests involved small, different groups and a limited prototype, further testing is needed before drawing wider conclusions. The project shows how watching users complete real tasks can guide practical improvements to an interface.

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Chapter 1

Introduction

1.1 Background

Music, podcasts and audiobooks can form part of the same person's listening activities, while being accessed through different applications. In such a situation, finding audio involves choosing both what to listen to and where to look for it. A listener may save an item in one application, continue a different item elsewhere and later need to remember which application holds each one. These possible organisational difficulties motivate the project.

Finding content is only one part of this problem. Saving an interesting item should also make it possible to find that item again. Similarly, returning to an unfinished podcast or audiobook requires a recognisable route back to the content and its listening position. For example, a student who stops listening to an audiobook during a study break should be able to identify the unfinished book and continue it later.

Combining the three audio types could provide a common place for these activities, but it also raises a design question. A mixed collection may be difficult to navigate if its content types are unclear or if too many items compete for attention. Music tracks, podcast episodes and audiobook chapters also need to remain identifiable within a shared interface. Common navigation and saving controls therefore have to be balanced with clear categories and information about each item.

Existing services already take different approaches to this organisation. Spotify combines music, podcasts and audiobooks in one service, while keeping the content types identifiable within its listening experience [1]. Apple offers separate applications for music, podcasts and audiobooks through Apple Music, Apple Podcasts and Apple Books [2]. These examples show that combining audio content is not a new idea, and neither approach is automatically easier for every user. Omnisound instead explores a specific interface question: whether shared search, a common library and clear content categories can make it easier to find saved material and continue unfinished listening without confusing the different audio types.

Omnisound is the project's response to this question: a web-interface prototype for finding, saving and resuming audio in one organised place. It primarily targets students

who regularly listen to music, podcasts or audiobooks, including listeners who use several content types and those who mainly listen to music. The project investigates how the interface can help these listeners keep track of saved and unfinished content.

The initial high-fidelity prototype (V1) is designed primarily for a desktop browser. It provides a shared search, content-type filters, a library of saved items and a Continue Listening area. Its two main task flows are finding and saving an item, then locating it in the library, and resuming an unfinished podcast or audiobook. These flows give the project a specific interaction focus within the broader audio concept.

1.2 Purpose

The purpose of the project is to investigate how a unified audio interface can support finding, organising and resuming different types of audio content while keeping the experience clear and manageable for the intended users. The motivation is to explore whether a common search and library can reduce the effort of returning to wanted content without making the differences between music, podcasts and audiobooks harder to understand.

This investigation also provides an opportunity to apply user-centred design within the HCI course. The course project requires an iterative process and usability testing in at least one iteration [3, pp. 2–3]. For Omnisound, the purpose is therefore pursued through requirements, alternative designs, prototypes and evaluation. The desired qualities are ease of learning, clear organisation and useful feedback; whether the design supports these qualities must be assessed through the project work.

1.3 Goals

The first goal was to identify what students need from an application that brings music, podcasts and audiobooks together. We wanted to focus on the functions that matter most, such as finding audio, saving content for later and continuing something that was not finished. These needs would help us decide what to include in Omnisound and give us a way to evaluate the interface beyond how it looks.

Another goal was to explore different design ideas before developing the interactive prototypes. We considered different ways to organise audio categories and display content, such as separate categories or a mixed view with filters, and cards or lists. The early sketches helped us explore these ideas, while V1 and V2 showed how the design developed into an interface that users could interact with.

The final goal was to test how well people could use Omnisound. We wanted to see whether users could complete the main tasks, where they experienced difficulties and which parts of the interface needed improvement. The first usability test helped us decide what to change in V2, and the second test allowed us to compare the two versions. Together, the sketches, prototypes and usability tests helped us explain our design choices and what we learned from the project.

1.4 Delimitations

Omnisound is an interface prototype. The project focuses on navigation, content organisation and the visibility of actions and listening progress. It does not implement music-streaming infrastructure, integration with commercial audio services, payment or subscription processing, licensing systems, backend databases or production account management. The prototype is implemented only as a front end. Limiting the implementation in this way allows the design work to concentrate on the selected interactions within the course project.

Search uses a sample catalogue, and the player controls and displayed progress represent the intended listening experience. They are sufficient to explore how users move through the interface, recognise content and interpret feedback. They do not demonstrate reliable audio delivery, synchronisation between devices or the operation of a real streaming service. Conclusions must therefore concern interaction with the prototype rather than the technical performance of a complete product.

The task scope is also limited. Search and save, retrieval from the library and continuation of unfinished podcasts or audiobooks receive priority. Other possible activities, such as managing a production account or exploring a large commercial catalogue, are outside this evaluation focus. The primary context is desktop use, so the project cannot establish usability across all mobile devices or listening environments. Likewise, conclusions drawn from evaluation with the intended student group should be considered in relation to that group and the tasks tested.

Following the first usability test, a second version of Omnisound was developed to address the problems identified during evaluation. The changes focused on reducing the size of the Home banner, making Continue Listening easier to discover, and improving the clarity of the Save/Saved control. A second usability test was then conducted using V2 and the same three core tasks, allowing the group to compare the two iterations. The testing procedures and findings are presented in the Method and Results chapters. Since Omnisound remains an interface prototype, the conclusions concern the tested interactions rather than the technical performance of a complete streaming service.

1.5 Division of Work

The project was carried out by Obid Alrhman Aljabi, Yacoub Dawli and Sarshar Chavoshi Rad.

Yacoub Dawli was responsible for leading the interviews and moderating the usability tests. Obid Alrhman Aljabi and Sarshar Chavoshi Rad were responsible for documenting participant responses and observations. During usability testing, Obid focused on participant behaviour and comments, while Sarshar documented task completion times and navigation steps.

The group collaborated on analysing user needs, discussing requirements and evaluating different interface designs. The design process included low-fidelity sketches, two

high-fidelity prototypes and usability evaluations. Findings from the first evaluation informed the changes introduced in the second prototype.

The project findings and design decisions were discussed within the group and formed the basis for the report.

Chapter 2

Theory

2.1 User-Centred Design and Iteration

User-centred design begins with understanding the people who will use an interface, what they want to achieve and the circumstances in which they will use it. Rogers, Sharp and Preece explain that understanding people's activities and differences helps designers question assumptions about their needs [4, pp. 16–17]. A target group therefore provides a starting point for research, rather than a reason to assume that all its members have the same habits or expectations.

User needs and goals guide the requirements for an interface. Designers can then explore alternative ways to support those goals and represent the ideas in prototypes. Evaluation with intended users examines whether a proposed solution works as expected. A prototype is a candidate solution that can be tested by observing people using it [5, Sec. "What Is a Testable Prototype?"]. This makes evaluation part of design, rather than a check reserved for the finished product.

Iteration connects evaluation to further design decisions. Findings can lead to revised requirements, a different alternative or changes to an existing interface, followed by another evaluation [5, Sec. "Why Test a Prototype?"]. Simply producing several versions does not establish this connection. A meaningful iteration should explain which problem motivated a change and examine whether the problem remains. This principle is relevant to Omnisound's student audience and its two evaluated high-fidelity prototype iterations.

2.2 Usability and User Experience

ISO 9241-11:2018 relates usability to specified users, goals and contexts of use through effectiveness, efficiency and satisfaction. These definitions are reproduced in ISO/TR 25060:2023 [6, Sec. 3.1]. Usability concerns what people can achieve with a system in a particular situation; it cannot be established from appearance alone.

Effectiveness concerns accurate and complete achievement of a goal, while efficiency concerns the resources used in relation to that outcome [6, Secs. 3.1.7–3.1.8]. In an audio

interface, effectiveness could involve successfully saving an item and finding it again. Efficiency concerns the time and effort required to reach that outcome. Successful completion and a straightforward route are thus different aspects of the same task. Satisfaction concerns how well the experience of use meets the user's needs and expectations [6, Sec. 3.1.9].

Learnability and utility are additional useful HCI concepts. Learnability concerns how easily a person can learn to use an interface, including getting started with unfamiliar controls. Utility concerns whether the available functions support what users need to do [4, p. 21]. For example, a clearly labelled search may be easy to learn, while a way to return to unfinished audio addresses a functional need. These concepts complement the three ISO dimensions; they are not additional components of the ISO definition of usability.

User experience includes people's perceptions and responses before, during and after interaction, including anticipated use [6, Sec. 3.1.15]. Satisfaction and frustration are relevant examples. Task completion alone cannot explain how an interaction feels. Together, these concepts provide a basis for examining searching, saving, retrieving and resuming audio without treating one measure as a complete assessment of the interface.

2.3 Interface Design Principles

Visibility means making the options needed for a task apparent to the user [4, p. 28]. An available function has limited value if people cannot discover it. In an audio interface, recognisable navigation and information about listening progress can help users identify where to go and what can be continued.

Feedback communicates the outcome of an action and the current system state [7, Sec. 1]. After saving an item, a visible confirmation can help the user understand that the action succeeded. Feedback should be timely and clear enough to support the next decision, such as continuing to browse or opening the saved collection.

Consistency means using similar terms, controls and behaviour for similar interactions, both within a product and in relation to familiar conventions [7, Sec. 4]. A shared saving action across different audio types can support predictable interaction. Consistency still allows relevant differences in content, such as episode titles and audiobook chapters.

Affordances concern the actions possible in the relationship between a person and an object. Signifiers are perceivable cues that communicate what can be done [8]. In a graphical interface, a button label, icon or visual treatment can signify an available action. A control's appearance and the action it actually supports should agree. This distinction helps explain why making an action available and making it understandable are separate design concerns.

2.4 Cognition and Mental Models

A mental model is a user's understanding of how a system works. It supports predictions about what an action will do and develops through previous experience [9, Sec. "What Are Mental Models?"]. Users and designers may have different models of the same interface. Familiarity with another audio service may, for example, shape expectations about where saved items belong or whether opening unfinished content will resume it. Clear category names and navigation can help communicate how a shared audio collection is organised.

Attention concerns which information people notice while pursuing a goal. Layout and visual prominence matter because competing information can reduce the visibility of relevant content [7, Sec. 8]. A large introductory area and a route back to unfinished listening may compete for attention on the same screen. This provides a reason to examine the relative prominence of elements, without assuming that size alone determines what every user notices.

Recognition over recall means displaying options and information that users can identify, rather than requiring them to remember details [7, Sec. 6]. Visible content categories, item titles and progress information can support this recognition. For an audio collection, these cues may help someone distinguish saved material from unfinished listening and choose a suitable next action.

2.5 User Research and Evaluation Methods

Qualitative interviews explore people's experiences, needs and expectations. Open questions and follow-up questions allow participants to explain what matters to them in their own terms [10]. Interviews can inform requirements, but they collect accounts of behaviour rather than direct observations of interface use. Combining these accounts with observations can provide a fuller basis for understanding users.

Affinity diagramming organises related observations or ideas into groups that express common themes [11, Sec. "Defining Affinity Diagramming"]. It can help a design team make sense of interview notes and identify connections between needs. The grouping is an interpretation of the material, so observations that differ from a common theme should also remain visible in the analysis.

Personas are research-informed representations of users' goals, needs and expectations. Qualitative personas draw on patterns in research to help a team focus its design decisions [12]. They summarise relevant differences between users rather than reproducing every interview as a separate profile. A persona based only on assumptions should not be treated as evidence about the intended audience, and a qualitative persona does not establish how common a user type is in the wider population.

Low-fidelity prototypes, such as paper sketches, represent early ideas with limited detail. They allow designers to explore layouts and navigation before investing in a detailed interface, and can be changed or discarded readily [5, Sec. "Benefits of Low-Fidelity Prototypes"]. More detailed, interactive prototypes allow evaluation of specific

controls and task flows [5, Sec. “Benefits of High-Fidelity Prototypes”]. The appropriate level of detail depends on the question being investigated; visual polish alone does not establish that a design meets users’ needs.

Think-aloud usability testing asks participants to express their thoughts while completing representative tasks with an interface [13, Sec. “Defining Thinking Aloud Testing”]. Observations of actions, hesitation and comments can reveal difficulties and help explain why a participant chose a particular route. These observations can inform redesign, including changes to the visibility or interpretation of controls.

Qualitative usability tests also have limitations. Speaking aloud creates an unusual situation, and prompts can influence behaviour [13, Sec. “Downsides of Think-Aloud”]. Findings from a small group can identify problems in the tested interactions, but do not automatically provide statistically generalisable conclusions. Task times and navigation steps should therefore be interpreted alongside observations and the test context, rather than treated as universal measures of performance.

2.6 Research Ethics

User research should respect participants’ ability to make an informed and voluntary decision. Vetenskapsrådet’s *God forskningssed 2024* explains that people should understand what participation involves, how information will be handled and how to withdraw consent [14, pp. 62–63]. Information should be understandable to the intended participants, who should have an opportunity to ask questions and decline or stop participation without pressure.

Privacy requires responsible handling of participant information. Only information needed for the study should be collected, and it should be protected appropriately. Removing names alone may not prevent identification when other details can be combined [14, pp. 66–67]. This matters when selecting background details, quotations and observations for a report.

Accurate reporting is also an ethical responsibility. Observations and measurements should be presented honestly, with interpretations and limitations made clear [14, pp. 11–12]. Results that challenge a design should be retained rather than adjusted to support a preferred conclusion. These principles provide the basis for describing the group’s documented ethical procedures in the Method chapter.

Chapter 3

Method

3.1 Overall Design Process

The project combined early concept development, design alternatives, prototyping and two rounds of usability testing. The concept and paper sketches established the intended listening activities and possible ways to organise the interface. These ideas were developed into the first high-fidelity prototype, V1. Its evaluation provided a basis for a targeted redesign, which produced V2. A second usability test then examined the updated interface using the same core task goals. Figure 3.1 summarises this sequence.

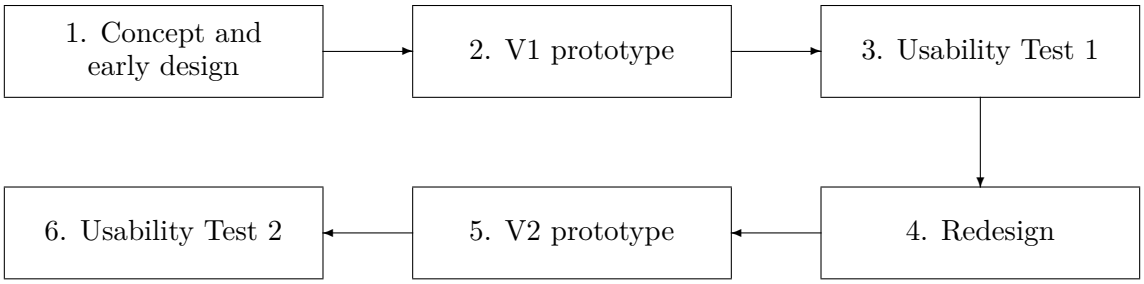


Figure 3.1: Omnisound's design and evaluation process (original project diagram).

This approach was suitable because combining audio types raises questions about how people understand categories, saved content and listening progress. An interactive prototype allowed the group to observe those decisions during use. The user-centred approach described in Section 2.1 therefore informed evaluation of concrete tasks. Iteration connected the observed problems to changes and another evaluation; creating a second version alone would not have provided that connection.

3.2 Initial Design Studio

The early design work was documented through concept material and low-fidelity sketches. The Session 4 concept sheet set out a shared audio interface for students, with finding,

saving and returning to content as its main activities. It also recorded alternative approaches to organisation: separate category tabs compared with a mixed feed and filters, and list views compared with cards. These alternatives framed the question of how to combine content while keeping its different types recognisable.

The original paper sketches explored a navigation area, grouped audio categories and a player layout. Together, they made it possible to examine where navigation, content and playback controls could fit before committing to a detailed interface. They provided concrete material for discussing interface ideas, without requiring each alternative to be fully implemented.

At this stage, the focus was the organisation of activities and information. The concept material connected possible layouts to shared search, a common library and continued listening. Keeping the drawings simple supported consideration of these relationships before visual detail and interactive behaviour became the main focus. The sketches and selected design are presented in the Design chapter.

3.3 Data Collection

To better understand what students might want from a combined audio application, the group prepared an interview plan during Project Session 2. The questions focused on which audio applications people normally used, what they liked about them, what could be improved and what they would expect from an application that brings music, podcasts and audiobooks together. The plan also included showing early interface ideas and asking about first impressions, missing features and how users would expect to complete common tasks.

Yacoub Dawli was assigned to lead the interviews, while Obid Alrhman Aljabi and Sarshar Chavoshi Rad were assigned to take handwritten notes. The questions were mostly open-ended so that participants could explain their habits and preferences in their own words. This approach was useful because the group wanted to understand what people expected from an audio interface before deciding how it should be organised [10].

During the V1 and V2 usability tests, we also asked participants about their listening habits. We wanted to know which applications they normally used, whether they listened mainly to music, podcasts or audiobooks, and which devices they preferred. Their answers gave us useful background information about the participants. Together with their comments and our observations during the tasks, this helped us understand how they experienced Omnisound.

3.4 Data Analysis and Requirements

The initial Omnisound concept and the chosen prototype tasks helped the group decide which functions were most important. The interface needed to let users find audio content, save items for later, return to their saved collection and continue listening to something

they had not finished. It was also important that users could easily tell the difference between music, podcasts and audiobooks, even though all three were available in the same application.

Rather than trying to build every feature found in a commercial audio service, the group focused on a few tasks that users could complete from beginning to end. For example, finding an artist was connected to saving that content and checking that it appeared in the Library. Resuming an audiobook required users to find unfinished content and return to the previous listening position. Browsing podcasts was another simple way to examine whether the different audio categories were easy to find. These tasks helped determine which screens, controls and interactions the prototypes needed.

The group also had to decide what could reasonably be included within the scope of a university project. Features such as subscriptions, account management and real audio streaming were left outside the prototype because the main focus was the interface and how people interacted with it. The later usability tests helped identify which parts of the interface were clear and which needed improvement. Although personas and affinity diagrams are discussed as HCI methods in Chapter 2, no completed outputs from those methods are documented in the project material, so they are not presented as steps used in the requirements analysis.

3.4.1 User Goals

The user goals describe what people should be able to do with Omnisound. Based on the project concept and the main activities selected for the prototype, we focused on four goals: finding audio content, saving something for later, returning to unfinished listening and browsing different audio categories without confusion.

For example, someone should be able to find an artist, save it and then locate it again in Library. Similarly, someone who has not finished an audiobook should be able to find it and continue from the previous position. These goals helped us decide which functions and screens to include in the prototype.

3.4.2 Usability Goals

Alongside the user goals, we wanted the main interactions to be easy to understand and complete. We focused on effectiveness, efficiency and learnability. This meant examining whether users could complete the tasks, how much time and navigation they needed, and whether they understood the interface without being shown which buttons to use.

We also wanted actions to provide clear feedback. For example, when someone saves an item, the interface should make it clear that the item has been saved and where it can be found again. During usability testing, we examined these goals through task completion, completion times, navigation routes, hesitation, participant comments and the need for assistance.

3.5 Prototyping Methods

Low-fidelity sketches and high-fidelity prototypes served different purposes. The paper drawings explored layouts and relationships between navigation, audio categories and playback. They allowed interface ideas to be considered before the group invested in detailed screens. This use of simple drawings is consistent with the value of low-fidelity prototypes for making design changes with limited preparation [5, Sec. "Benefits of Low-Fidelity Prototypes"].

V1 was a desktop-first, browser-based prototype implemented as a front end. Its main screens were Home, Search, Library, content detail and the full player. These screens supported connected task paths rather than only showing isolated layouts. Users could search the sample catalogue, open an item, change its saved state, find it in the Library and open unfinished spoken audio at its displayed progress. This level of interaction allowed the tests to examine labels, navigation choices, controls and feedback.

The catalogue and listening states represented the intended experience. The prototype did not provide a complete streaming service, so opening the player at an existing position was an interface response, not a test of audio delivery. V2 retained this implementation scope and the main task paths. It provided a revised interface for a second evaluation, while V1 remained available as the earlier version. Detailed screen descriptions and visual comparisons belong in the Design chapter.

3.6 Usability Testing

Two usability-testing rounds were conducted with three participants in each round. P1, P2 and P3 tested V1, while P4, P5 and P6 tested V2. All six were students, aged 20–26, who reported regular use of audio applications. Their backgrounds included mainly music listening and combinations of music, podcasts and audiobooks. These characteristics were relevant to the intended student audience. The records describe participants' listening habits but do not specify how they were recruited.

The evaluations used the interactive prototypes in a browser on a laptop, within the project's desktop-use focus. The V1 plan identifies Mittuniversitetet in Sundsvall as the location and specifies that the prototype ran locally. The V2 plan also specifies a laptop and browser. Participants' usual mobile or computer use was recorded as background information; the sessions did not constitute a comparison between devices.

The group divided moderation and observation between its members. Yacoub Dawli led the sessions and presented the tasks. Obid Alrhman Aljabi documented participant behaviour and comments, while Sarshar Chavoshi Rad recorded task times, clicks and navigation observations. This division allowed the moderator to attend to the participant while the observers recorded the interaction.

Each session began with an introduction to the university project and the purpose of the test. After information about participation and consent, the moderator asked which audio applications the participant normally used, which audio types they listened to, how

often they listened and whether they normally used mobile devices, computers or both. The three tasks were then presented individually. Participants were asked to explain what they were looking for, what they expected to happen and anything they found unclear while using the interface. This followed the think-aloud approach introduced in Section 2.5 [13, Sec. “Defining Thinking Aloud Testing”].

The instructions described the intended goal without specifying the control to select. For example, the unfinished-audiobook task did not tell participants to use Continue Listening. This left their choice of route available for observation. When prompting was needed, the moderator asked neutral questions about what the participant was thinking or expected. Participants were encouraged to try what made sense to them instead of being immediately directed to a solution.

Table 3.1 summarises the three core tasks and the completion criteria recorded in the test documentation. The first task included checking that the saved item could be found again. The second concerned returning to an existing listening position, while the third examined access to one content category. These tasks covered the main interaction priorities without requiring every prototype function to be tested.

Table 3.1: Core task goals and completion criteria used in both testing rounds.

Task	**Goal**	**Completion criterion**
1	Find and save Weezer for later.	The item is saved and can be found in the Library.
2	Find an unfinished audiobook and continue listening.	The player opens at the existing progress position.
3	Find where to browse podcasts.	The Podcasts category is reached.

The same core goals and completion criteria were used in the second round so that the group could compare the two versions without replacing the activities being examined. The wording allowed different valid routes to the goal. Finding content through a category, for example, could still meet the first task’s criterion. The records could therefore distinguish task completion from the particular route chosen.

The task records covered completion, elapsed time in seconds, click counts and navigation paths, alongside notes on hesitation, incorrect actions and any need for help. Relevant think-aloud comments accompanied these observations. Completion related to effectiveness, while time, clicks and routes provided information about effort and efficiency. Hesitation and comments helped examine whether controls were discoverable and whether their feedback was understood. These records supported interpretation of the measured values; a time or click count alone did not explain a difficulty.

Post-test questions covered the overall impression, easy or confusing parts, unclear labels, missing functions and suggested changes. The V2 records also included questions about Save and discoverability. The group reviewed these responses alongside the task observations in each round. This connected participants’ impressions to what happened during interaction. Individual accounts, measurements and the comparison of results are

presented in the Results chapter.

3.7 Iterative Evaluation

After the first evaluation, the group brought together task records, navigation paths and post-test comments to identify issues for redesign. The V1 analysis linked each identified issue to supporting observations, its effect on the tested task and a proposed action. The prioritised problem list distinguished concerns about reaching or discovering task-related content from smaller concerns about visual clarity. It also identified parts of the interface to retain.

The resulting redesign focused on Continue Listening visibility, the height of the Home introductory area and the clarity of the Save/Saved control. These areas connected the recorded concerns to specific changes that could be examined in another test. The basic navigation, categories, Library and core task goals were kept consistent. This limited the scope of the redesign and retained the main interaction context for comparison.

V2 was then evaluated with new participants using the procedure and tasks described in Section 3.6. Using different participants avoided repeating the tasks with people who had already learned the V1 paths. The second-round analysis considered the same completion criteria, measures, routes and participant feedback, including observations related to the targeted changes.

This comparison concerned the tested prototype interactions. With three different participants per round, differences between the records cannot be attributed solely to the redesign or generalised to all students. The purpose was to examine the revised design against the earlier concerns, without assuming that changing the interface established an improvement.

3.8 Research Ethics

The participant introductions explained that the group was evaluating the interface, not the person using it. They described the university project, the observation and written note-taking, and the intended use of the information. Participants were told that taking part was voluntary, that they could stop at any time and that they could raise questions or discomfort. The moderator then asked whether they were comfortable participating, and the records include agreement before the tasks. This approach addressed informed and voluntary participation as discussed in Section 2.6 and in Vetenskapsrådet's guidance [14, pp. 62–63].

The introductions stated that written notes would be anonymous and used for the university project. The retained records identify participants as P1–P6 instead of using personal names. Background information about listening habits and device use provided context for the tests without requiring names in the report.

Accurate reporting also informed the treatment of the material. The testing records

distinguish observations and participant comments from the group's interpretations. This distinction is maintained when describing the method and reporting the results, consistent with the responsibility to communicate research honestly [14, pp. 11–12].

Chapter 4

Design

4.1 Initial Design Alternatives

The early concept considered separate audio categories and a mixed feed with filters. Categories give listeners a clear starting point when they already know which audio type they want. A mixed feed supports browsing different types together, but needs clear labels and filters to distinguish them.

Cards and lists offered another choice. Cards leave room for artwork and visual recognition, but fewer items fit on the screen. Lists show titles, creators and other information more compactly. Their suitability depends on whether the user is browsing or looking for a particular item.

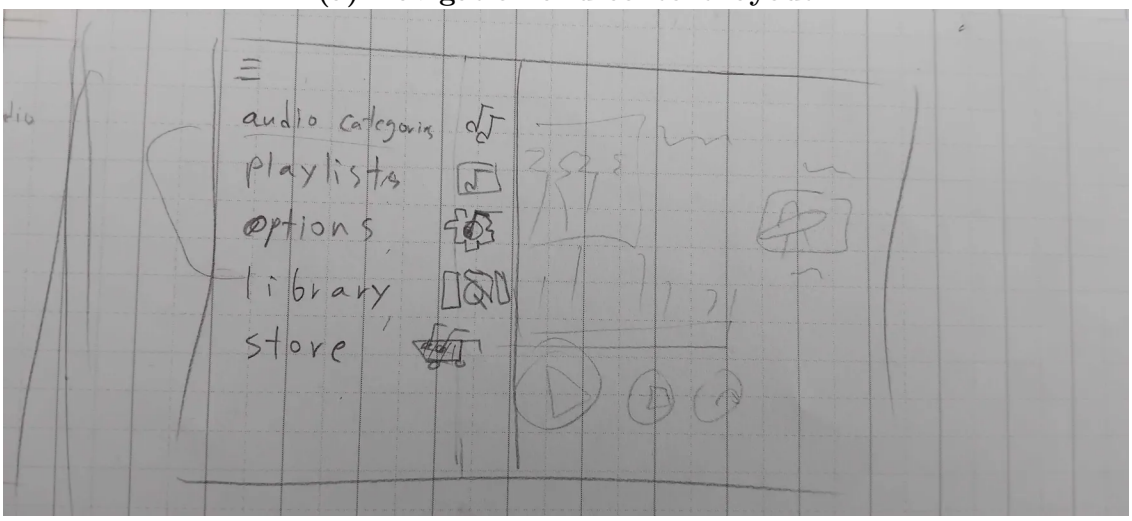
The chosen interface combines these ideas: cards in separate Home sections, a list of search results and one Library with content filters. Each audio category has a browsing route, while saving follows a shared pattern. This gave us a starting design that we could test and improve.

4.2 Lo-Fi Sketches

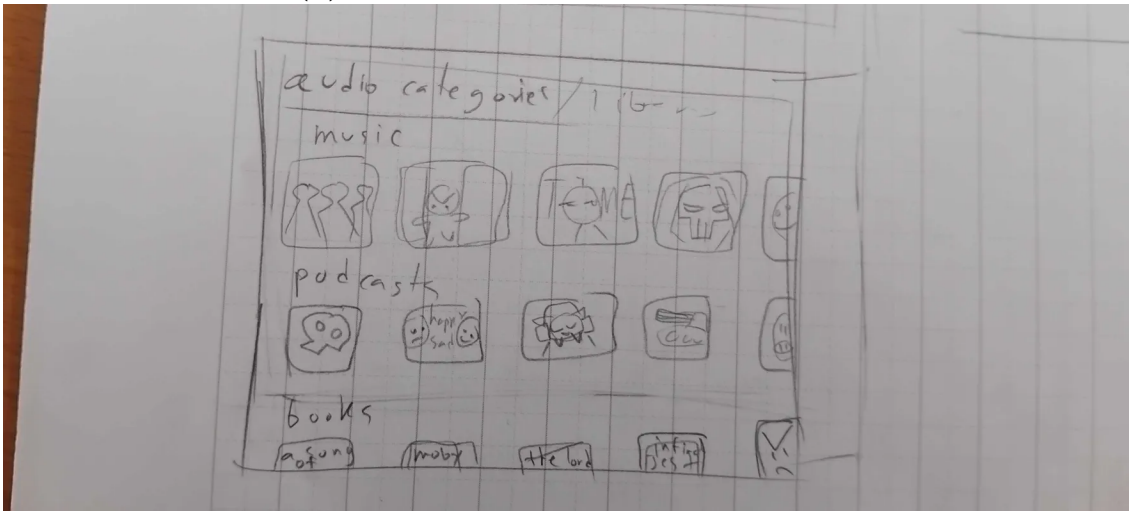
Figure 4.1 presents the group's three original pencil sketches. Sketch (a) places a labelled navigation area on the left, with entries for audio categories, playlists, options, Library and a store. The area on the right contains rough content blocks and playback symbols. Sketch (b) concentrates on browsing: rows labelled music, podcasts and books contain small drawings of covers. Sketch (c) explores a player around Weezer, with central artwork, a large play symbol, neighbouring controls and a horizontal volume control below.

These drawings made the layout ideas visible before the interactive screens were built. They helped the group consider how much space navigation, content and playback would need. Their rough appearance also left room to change the arrangement without first producing detailed visual designs. They were starting points rather than specifications: for example, the store shown in sketch (a) was not included in the tested prototype.

(a) Navigation and content layout



(b) Audio categories and cover cards



(c) Player layout

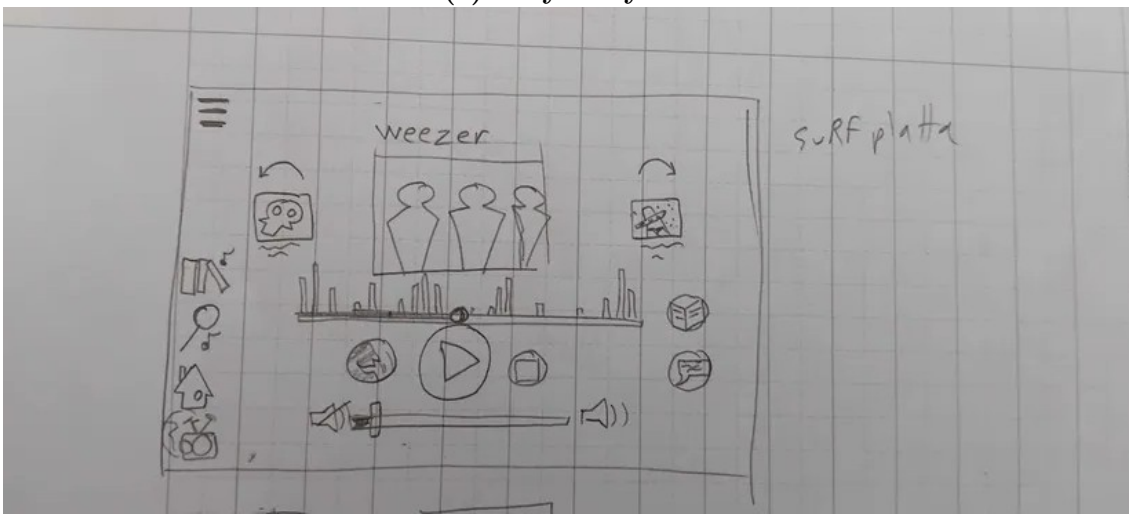


Figure 4.1: The three original project sketches, exploring navigation, category rows and playback. The photographs retain their original content and proportions.

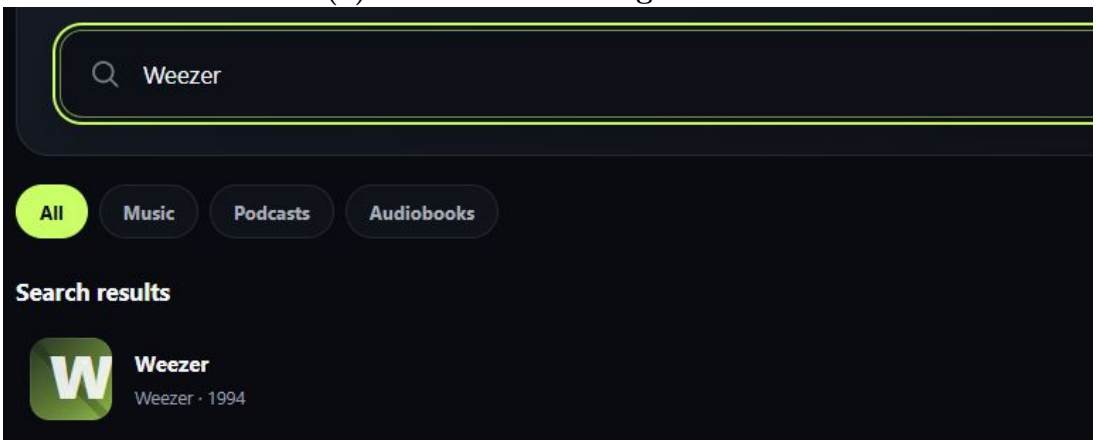
4.3 Omnisound Interface Structure

Omnisound has five main screen types: Home, Search, Library, content detail and the full player. Persistent navigation provides Home, Search and Library, followed by separate Music, Podcasts and Audiobooks routes. These remain available when opening content or playback.

Home includes Continue Listening for unfinished spoken audio. Search covers the sample catalogue, with type filters. Library collects saved items and offers similar filters. Details provide Save and Play or Resume. The full player groups listening controls; a smaller player remains available during browsing after playback opens.

Figure 4.2 shows Weezer in Search and, after saving, in Library. Results use lists for scanning, while Library uses covers and titles for recognition. Shared destinations and controls connect the audio types without requiring different saving and playback patterns.

(a) V1 Search: finding Weezer



(b) V1 Library: Weezer after saving

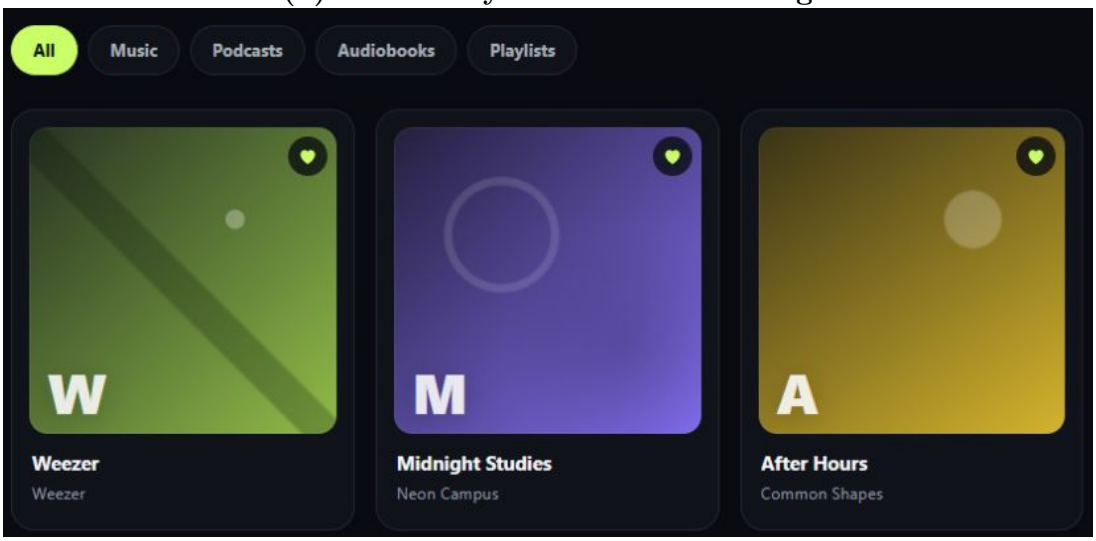


Figure 4.2: V1 Search and Library, cropped to show the query, filters, result and saved collection.

4.4 Hi-Fi Prototype V1

V1 developed the sketches into the first interactive high-fidelity prototype. Instead of viewing isolated drawings, users could follow the main activities from one screen to another. The category rows and artwork remained part of the design, while labelled buttons and visible states made searching, saving and resuming available for testing.

The Home screen in Figure 4.3 starts with a large introductory banner. Below it, a small Continue Listening label sits above the heading "Back in progress". The unfinished items show their audio type, title, creator and listening progress. Separate Music, Podcasts and Audiobooks sections follow further down the page. The sidebar provides a direct route to each category even when its Home section is below the visible area.

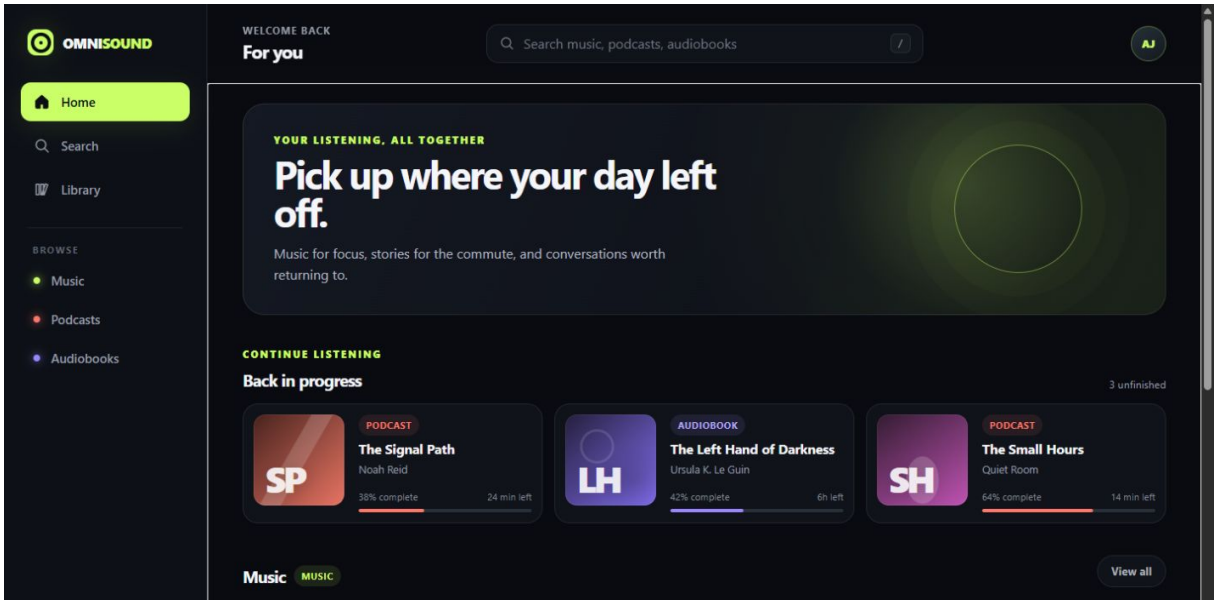
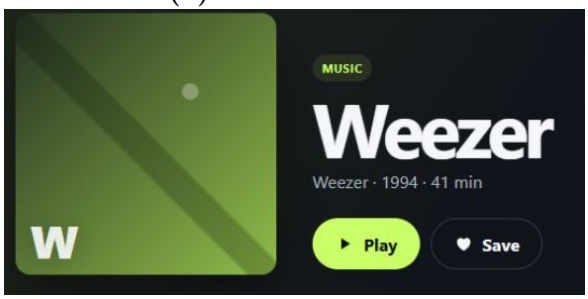


Figure 4.3: The V1 Home screen, showing the persistent navigation, large introductory banner and unfinished content. The capture shows the upper part of Home; further category rows continue below.

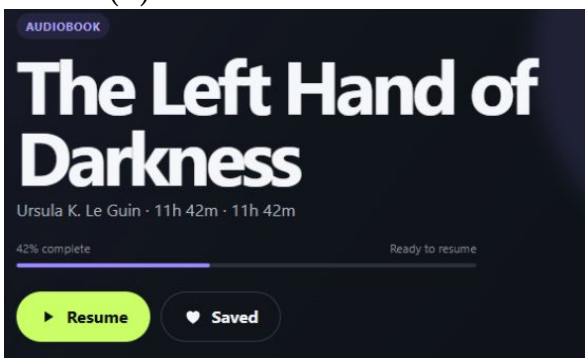
One important workflow begins by searching for Weezer and opening the result. Its detail screen places Save beside Play, as shown in Figure 4.4(a). Selecting Save changes the label to Saved and briefly displays "Weezer saved to your Library". Opening Library then shows the saved item, as in Figure 4.2(b). The connection matters because saving should provide a route back to the content, rather than only changing a button on the current screen.

The other main workflow returns to unfinished audio. A user can open *The Left Hand of Darkness* from Continue Listening and recognise its existing progress on the detail screen. Resume then opens the full player at that displayed position. Figure 4.4(b–c) shows these connected states. The player includes progress, play/pause, volume and spoken-audio speed controls. Its sample content and progress represent interface states; they do not demonstrate a complete streaming service.

(a) Weezer detail



(b) Unfinished audiobook



(c) Full player after Resume



Figure 4.4: V1 saving and resuming workflows: Save beside Play on the Weezer detail page, progress and Resume on the audiobook detail page, and the resulting full-player view. Crops focus on the relevant content and controls.

4.5 Design Decisions and HCI Connections

The principles in Section 2.3 guided concrete parts of the interface. The sidebar labels and categories in Figure 4.3 support visibility. Labels such as Music and Audiobook explain what an item is before it is opened, without relying only on artwork or colour.

Feedback connects actions to outcomes. Saving changes the label and displays a confirmation naming Library. Navigation highlights show the current destination, while play/pause controls reflect the player state. However, clear feedback after saving does not ensure that Save is easy to find beforehand, as the first test showed.

Consistency links the audio types. Their cards open similarly arranged detail pages, with Save beside Play or Resume. The lists below contain tracks, episodes or chapters as appropriate, while the main interaction pattern stays familiar. This connects expectations and behaviour as discussed in Section 2.4.

Recognition also shaped the screens. Titles and covers help users identify Library items, and progress bars identify unfinished audio. These cues reduce the need to remember exact titles or listening positions. Buttons such as Play, Resume and Save use familiar icons and clear labels to show users what they do. Play and Resume state what will happen; the heart with Save gives the icon a written meaning. The redesign addressed cues that needed more emphasis.

4.6 Redesign Decisions

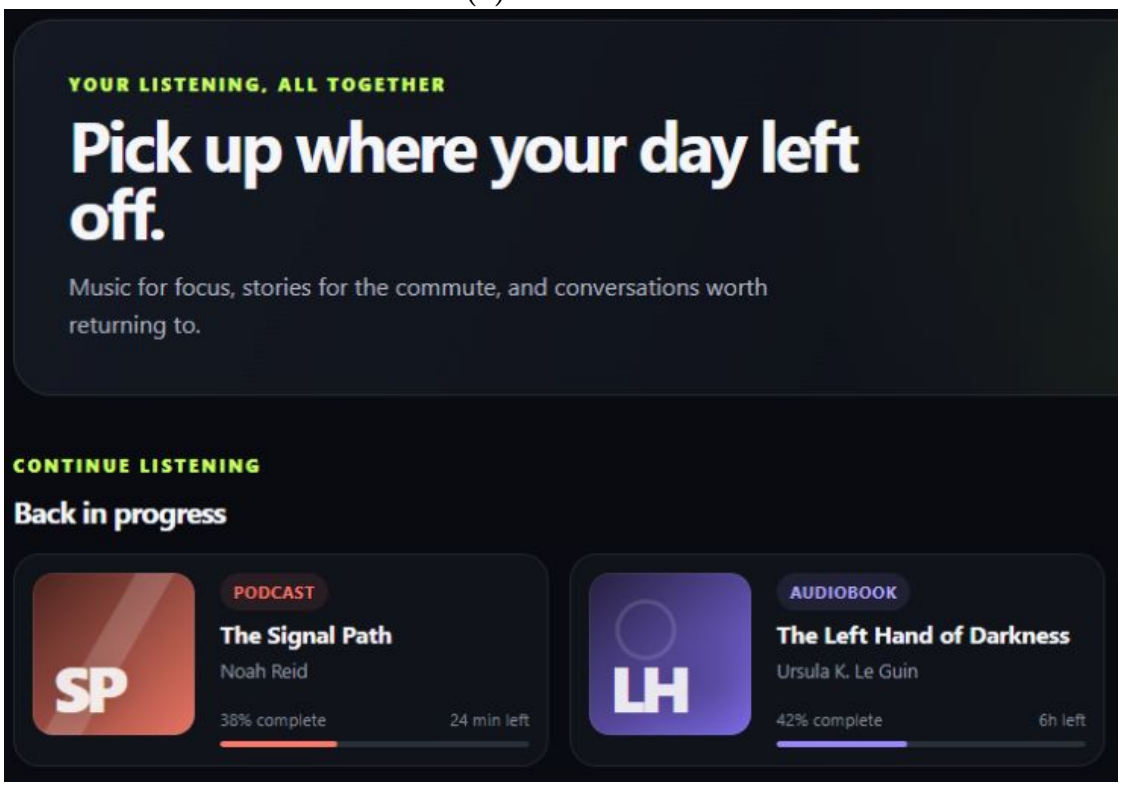
The first usability test showed that some participants looked for unfinished audio through Audiobooks or Library instead of Continue Listening. Those routes worked, but the intended Home shortcut was being overlooked. The group therefore focused the redesign on making that section easier to notice, reducing the space taken by the introduction and clarifying the saving control.

Figure 4.5 shows the Home changes. V2 reduces the banner's height and the space before unfinished content. It also replaces the small Continue Listening label and "Back in progress" heading with a single, larger Continue Listening heading and a lime marker. The unfinished cards and their progress information remain. The intention was to bring attention to content that users could return to, while keeping a short introduction to Omnisound.

The Save control received a smaller change. The first test recorded a brief hesitation in locating it beside Play, so V2 strengthened its border and gave the unsaved and saved states more distinct appearances. The label and confirmation message were retained because they explained the action's result. Figure 4.6 shows the revised states.

The group kept the sidebar, Search, Library and audio categories because the first evaluation supported these routes. Library also provided another way to reach unfinished content. We then tested V2 to see whether these changes made the interface easier to use.

(a) V1 Home



(b) V2 Home

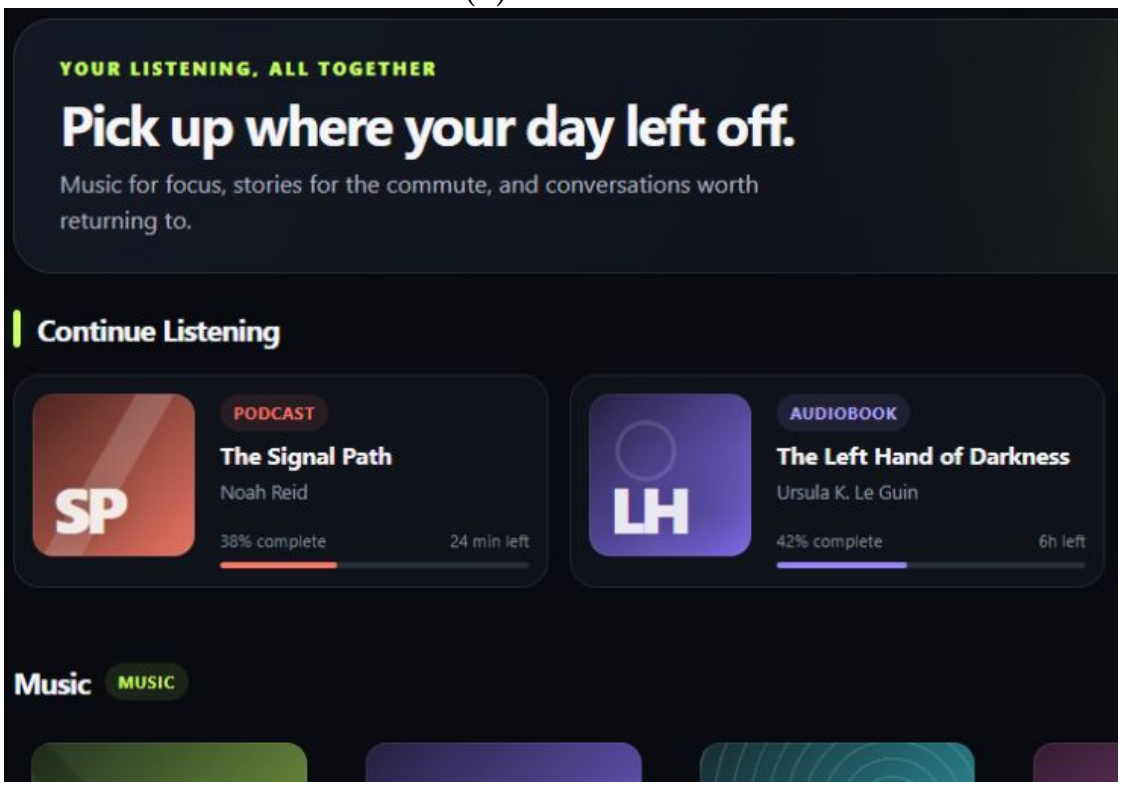


Figure 4.5: Home before and after the redesign. Both prototypes were captured at a 1440×900 viewport and 100% zoom, then cropped to the same region of the banner and first two unfinished cards, at the same scale. The smaller banner and revised heading bring unfinished content higher on V2 Home.

4.7 Hi-Fi Prototype V2

V2 keeps the five main screen types and navigation structure of V1. The Home comparison in Figure 4.5 also shows the final arrangement: the introduction occupies less space, and the more direct Continue Listening heading gives unfinished content greater emphasis. Users can still search, browse categories or open Library through the same destinations.

On the detail page, the unsaved control uses an outlined heart and the label Save. After saving, it shows a filled heart, lime styling and Saved. The comparison in Figure 4.6 captures the same Weezer page before and after selecting Save. The control stays beside Play, so its position remains familiar while its changed appearance communicates the current state. The brief Library confirmation is also retained, giving immediate feedback as well as a state that remains visible on the page.

(a) Before saving **(b) After saving**

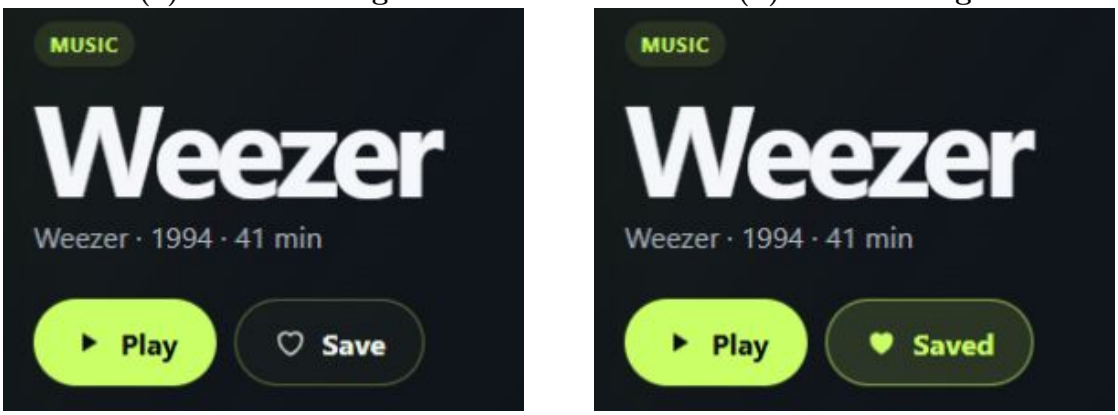


Figure 4.6: The Save and Saved states in V2, captured before and after saving the same item. Matching crops preserve the detail context; only the saving control changes. The temporary confirmation has disappeared in the second capture.

The group evaluated V2 in a second usability test using the same core tasks: finding and saving content, resuming unfinished audio and browsing podcasts. This completed the sequence from V1 evaluation to redesign and another test. Chapter 5 presents the recorded outcomes and comparison between the two rounds.

Chapter 5

Results

5.1 Data Collection Results

Six students took part in the usability tests: P1–P3 used V1 and P4–P6 used V2. All six reported using audio applications every day, and Spotify appeared in every participant's account. Other applications included YouTube, YouTube Music, Audible, Pocket Casts, Apple Podcasts and Storytel. Table 5.1 summarises their reported preferences, applications and usual devices.

Music was part of all six participants' listening habits. P1 and P4 also listened to podcasts, while P3 described podcast listening as rare. P2, P5 and P6 mainly listened to music and audiobooks. The group therefore included both mainly music listeners and people who regularly combined audio types.

Table 5.1: Reported audio preferences, applications and usual devices.

ID	**Round**	**Audio preferences**	**Usual applications**	**Usual devices**
P1	V1	Mainly music; some podcasts	Spotify, YouTube	iPhone, laptop
P2	V1	Music and audiobooks	Spotify, Audible, Pocket Casts	Android phone, Windows PC
P3	V1	Mainly music; rare podcasts	YouTube Music, Spotify	Android phone
P4	V2	Mainly music; some podcasts	Spotify, YouTube Music	Android phone, Windows laptop
P5	V2	Music and audiobooks	Spotify, Apple Podcasts, Storytel	iPhone, MacBook
P6	V2	Music and audiobooks	Spotify, Audible	Android phone, Windows laptop

Phone use was common, but several participants also listened on computers. P1 and P4 mentioned computer use while studying. P2 used both platforms for music and mainly a phone for audiobooks. P3, P5 and P6 mainly used phones.

5.2 Identified User Needs

The project concept already prioritised finding content, saving it in a shared Library and returning to unfinished listening. Keeping music, podcasts and audiobooks recognisable was another starting requirement. The testing records provided observations about how participants used these functions and what they found unclear.

Finding and retrieving content involved more than one route. Participants used Search or Music to locate Weezer, then saved the item and checked Library. Their actions showed how the original finding-and-saving goal was used in practice. Library also served another purpose for P3, who looked there for the unfinished audiobook. Saved collections and unfinished listening therefore overlapped in this participant's route through V1.

The feedback also identified specific priorities within the concept. P2 and P3 did not use Continue Listening in the first test and later said they had not noticed it. P1 suggested making Save more obvious. In V2, participants recognised the changed Saved state and checked the item in Library. These observations concerned finding the relevant control and understanding what happened after selecting it. Category names were also used directly during the tasks, and participants' post-test comments described the separation between audio types as understandable.

5.3 Usability Test 1 Results

P1, P2 and P3 completed all three V1 tasks: finding and saving Weezer, resuming an unfinished audiobook and browsing podcasts. This gave nine completed tasks out of nine attempts, with no direct assistance from the test leader. Table 5.2 presents the recorded completion times and outcomes.

Table 5.2: V1 task times in seconds and completion results.

Task	**P1**	**P2**	**P3**	**Completed**	**Direct help**
Find and save Weezer	31	24	28	3/3	None
Resume audiobook	19	27	35	3/3	None
Browse podcasts	4	5	7	3/3	None

For the first task, P1 selected Search in the sidebar, while P2 opened Search through the top search field. P3 opened Music and found Weezer among the visible cards. Each participant opened its details, selected Save and found the item in Library. Their routes differed, but all reached the requested saved collection.

P1 noticed Play before Save and briefly paused before locating the saving control. After saving, P1 understood the confirmation and opened Library to check the result. P3 initially assumed the heart would add the item somewhere, then named Library after reading the confirmation message. In the post-test response, P1 suggested: "Maybe make Save slightly more obvious." No participant needed help to complete the saving task.

The audiobook task produced the largest spread in V1 times, from 19 to 35 seconds. P1 used Home, Continue Listening, the audiobook details and Resume. P2 instead opened Audiobooks from Home, selected *The Left Hand of Darkness* and chose Resume. P3 went from Home to Library, opened the same audiobook and selected Resume. The player opened at the existing progress in all three cases.

Only P1 used Continue Listening directly. P2 scanned the audiobook cards through the category route, while P3 passed the Continue Listening section and searched in Library. P3 said during the task: "Maybe unfinished things are saved here." The progress information and Resume control then identified the audiobook as unfinished. The recorded resume-task click counts were two for P1, three for P2 and four for P3.

All three participants reached Podcasts. P1 and P2 selected it directly in the sidebar. P3 accidentally selected Music first, noticed its active highlight and immediately switched to Podcasts. This was the recorded wrong action in that task; P3 still completed it without help.

Post-test comments matched the different audiobook routes. P2 said: "I didn't immediately notice Continue Listening. I went to Audiobooks instead." P3 also said the section had not been noticed and commented: "Maybe make Continue Listening more obvious. The big banner above it takes a lot of space." The main V1 observations were completed tasks through several routes, an overlooked Home shortcut for P2 and P3, a brief hesitation around Save for P1 and one category-selection error followed by immediate recovery.

5.4 Usability Test 2 Results

P4, P5 and P6 completed the same three core tasks using V2. All nine task attempts were completed without direct assistance. Their recorded times are shown in Table 5.3, using the same layout as the V1 results.

Table 5.3: V2 task times in seconds and completion results.

Task	**P4**	**P5**	**P6**	**Completed**	**Direct help**
Find and save Weezer	22	24	21	3/3	None
Resume audiobook	13	15	12	3/3	None
Browse podcasts	4	4	3	3/3	None

P4 used the top search field to find Weezer, and P5 selected Search in the sidebar. P6 opened Music and found the item directly. All three selected Save on the detail page,

recognised the Saved state and opened Library to check that Weezer appeared there. The music-category route therefore remained available alongside Search in the second round.

P4 noticed Save immediately and described it as obvious. P5 briefly looked between Play and Save before selecting it; the notes record a pause of approximately one second. After the button changed to Saved, P5 said: "Okay, now I know it worked." P6 selected Save without recorded hesitation.

All three participants used Continue Listening to resume the unfinished audiobook. Their route was Home, Continue Listening, the audiobook details and Resume. Each then reached the player at the existing position. P4 looked below the banner and selected the audiobook, P5 saw the section without scrolling, and P6 referred to the displayed progress when identifying unfinished content. The notes record two clicks and no hesitation for each participant on this task.

For podcast browsing, P4, P5 and P6 each selected Podcasts in the sidebar. The recorded times were four, four and three seconds respectively, with one click each. No wrong category selections or difficulties were recorded for this task in V2.

The post-test responses referred to both the targeted controls and the general navigation. P4 described the sidebar and searching as the easiest parts and said: "No, Continue Listening was pretty visible." P4 also said the heart and the word Save made the action clear. P5 named Continue Listening and navigation as the easiest parts, while P6 named categories and Continue Listening. All three described Save as understandable, and none reported a confusing label.

The feedback also included suggestions beyond the tested tasks. P5 wanted Recently Played as a possible later feature, and P6 suggested sorting the Library. Neither suggestion described an obstacle that prevented task completion. The V2 records therefore contain successful task outcomes, direct audiobook-resumption routes and a brief saving pause, alongside ideas for additional functions.

5.5 Comparison Between Iterations

Task completion was the same in both rounds: nine out of nine attempts, with no direct help required. The recorded differences concerned time, route choice and hesitation rather than additional completed tasks. V1 participants took between 24 and 31 seconds to find and save Weezer, compared with 21–24 seconds in V2. Audiobook resumption took 19–35 seconds in V1 and 12–15 seconds in V2. Podcast browsing took 4–7 seconds in V1 and 3–4 seconds in V2.

Figure 5.1 compares the average completion times. Each average uses the three participant times for that task in its round. The means are 27.7 seconds for finding and saving in V1 and 22.3 seconds in V2; 27.0 and 13.3 seconds for resuming the audiobook; and 5.3 and 3.7 seconds for browsing podcasts. Values are rounded to one decimal place.

The largest difference in average time was in audiobook resumption. Route choice also differed on that task: one of the three V1 participants used Continue Listening directly,

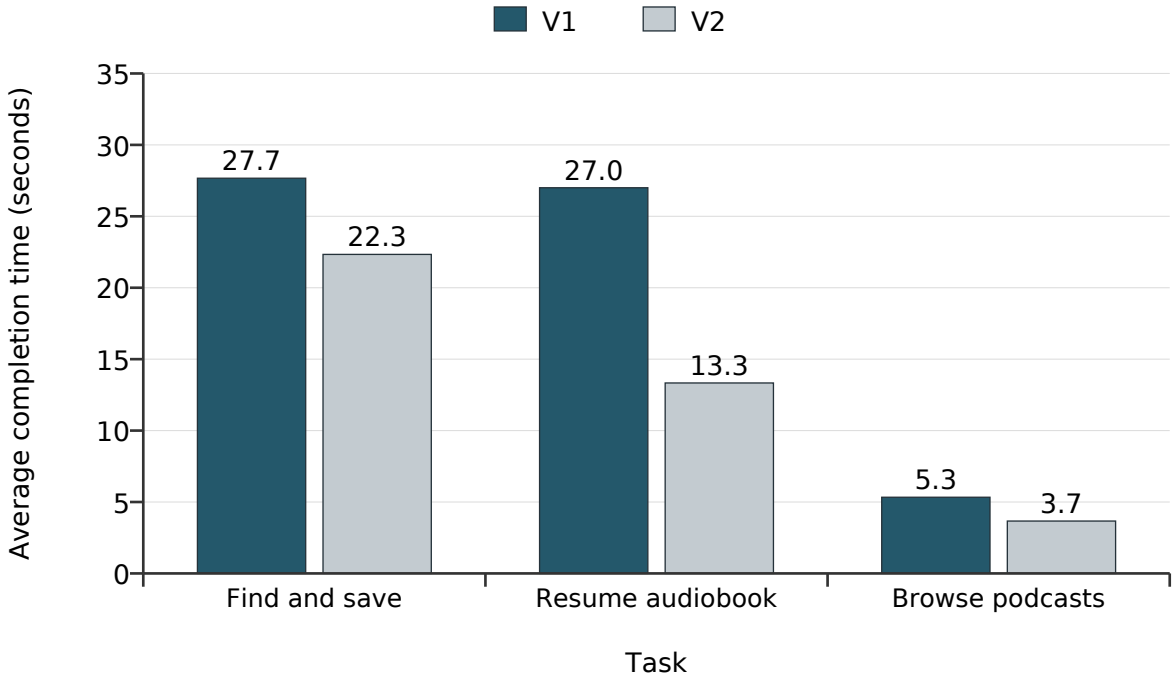


Figure 5.1: Average task completion times, based on three participants per testing round, with different participants in V1 and V2. Bar labels show means rounded to one decimal place.

compared with all three V2 participants. The other V1 participants still completed the task through Audiobooks or Library. Finding Weezer through Music occurred in both rounds, alongside the two entry points to Search.

The saving records include a hesitation for P1 in V1 and a shorter pause for P5 in V2. Both completed the task and checked Library. Category navigation also reached the requested destination in both rounds; P3's initial Music selection in V1 was followed by immediate recovery, while no such wrong selection was recorded in V2.

The recorded completion times were lower in the second round, especially for resuming the audiobook, and Continue Listening was used more often. These are comparisons between two groups of three different participants; the lower times alone do not show that the redesign caused the difference. Across both rounds, all participants completed the tested tasks without direct assistance.

Chapter 6

Discussion

6.1 Interpretation of Results

Both versions supported the main tasks for the participants who tested them. Tables 5.2 and 5.3 show nine out of nine completed tasks in each round, without direct assistance. This suggests that the tested navigation and core interactions were understandable. However, reaching a goal did not mean that every route was equally easy to find.

The clearest difference concerned the unfinished audiobook. Only one of the three V1 participants used Continue Listening directly; the others went through Audiobooks or Library. All three V2 participants used Continue Listening, and the average time fell from 27.0 to 13.3 seconds (Figure 5.1). The second group found unfinished content more directly and completed the task faster. The updated Home screen may therefore have made the intended shortcut easier to notice.

Saving showed a smaller change. P1 briefly hesitated before finding Save in V1. All V2 participants understood the Save/Saved state, although P5 briefly paused before saving. The clearer button appearance and retained confirmation may have helped communicate both the action and its result. The states shown in Figure 4.6 provide a visible reminder after the temporary confirmation disappears.

V1 already offered useful ways to complete the tasks. Search, Library and the categories supported several successful routes, including finding Weezer through Music. P3 also recovered immediately after selecting the wrong category. Keeping these routes while changing a few specific areas was a reasonable response to the findings. The remaining brief pause in V2 is a reason to keep observing the saving control, rather than assume that every small difficulty has disappeared.

6.2 Connection Between Theory and Findings

The Home findings connect visibility with attention, as discussed in Sections 2.3 and 2.4. Continue Listening was available in V1, but its presence alone did not make it an obvious starting point for P2 and P3. The large banner occupied space above it. Reducing the

banner and using a more direct heading gave unfinished content greater prominence, as Figure 4.5 shows. This offers a plausible explanation for the different routes in V2, without treating appearance as a complete explanation of behaviour.

Feedback mattered after an action as well. The changed Saved label and the message naming Library told participants what had happened and where the item could be found. P3's move to Library after reading the confirmation is a concrete example. The active navigation highlight also helped P3 recognise an accidental category selection. These responses supported the next decision instead of leaving the participant to guess the current state.

Consistency and recognition help explain why the alternative routes still worked. Category names, titles and progress information gave participants something to recognise. P3 expected unfinished content to be in Library and could identify the audiobook there. Shared controls across the audio types supported that route. Keeping the navigation and Library structure in V2 preserved these interaction patterns alongside the clearer Home shortcut.

The iteration also put the user-centred approach in Section 2.1 into practice. We chose the V2 changes from observed hesitation, route choices and comments. This gave each change a specific problem to address and a reason for another test. Personal preferences about the appearance could not have told us whether listeners would notice Continue Listening during a task.

6.3 Strengths and Limitations

A strength of the project was the connection between its stages. Early sketches led to interactive prototypes, and the first test led to focused changes that were tested again. The same core tasks gave both rounds a common basis for comparison. Retaining V1 also made the design changes visible. Together, the observations, comments, times and routes provide a clearer account of the interaction than a finished interface alone would.

Only three participants tested each version, so the findings describe a small set of experiences. Different people took part in the two rounds. This avoided repeating the tasks with participants who had already learned the V1 routes, but differences between people could also affect times and choices. We therefore cannot attribute the lower V2 times only to the redesign. The Home changes were introduced together, so the tests also do not separate the effect of the smaller banner from the clearer heading.

The tasks covered finding and saving content, resuming an audiobook and browsing podcasts. They did not examine every function or repeated use over time. Testing took place in a laptop/browser setting, although many participants normally listened on phones. The results therefore give little information about the full mobile experience. The sample catalogue and displayed progress also limited what could be tested: real audio delivery, large collections, subscriptions and account handling were outside the front-end prototype.

The initial interview plan is retained, but completed initial interview responses are not. This limits how closely the early requirements can be traced to interview findings. The concept and later testing still identify useful priorities for the prototype, but they provide a narrower view than a fuller record of students' needs would. Our first goal was therefore addressed within the selected listening activities, rather than through a complete account of everything students might need from an audio service.

6.4 Evaluation of Methods

Starting with sketches was useful for considering navigation, content and playback before building detailed screens. It left room to change the arrangement without first completing an interactive version. The later prototypes served a different purpose: participants could actually search, save, browse and resume content. We could observe the connection between screens and the result of an action, rather than rely only on opinions about static layouts.

Think-aloud comments helped explain choices that the completion totals would have hidden. P2 looked in Audiobooks and P3 expected unfinished content in Library. Their comments and routes showed why the intended Home shortcut needed attention. Task times and click counts added information about the effort involved, while the observations showed where participants paused or changed direction. Combining these records was more useful than treating completion or time as the whole result.

The division between moderation and observation also helped the group follow the interaction while recording it. Explaining voluntary participation and using participant labels also addressed consent and privacy, as described in Section 3.8. In later work, we would retain these practices and make the link between interview notes, requirements and design decisions easier to follow. A larger and more varied participant group, with consistent task wording and test conditions, would also give us a broader basis for reviewing the interface.

6.5 Future Work

The next evaluation should include mobile interaction and a wider range of listening habits. It should also ask people to return to saved or unfinished audio after a break or over several visits. This would examine whether Library and Continue Listening remain useful when users have to recognise content later, rather than immediately after a task. More realistic playback and retained progress across visits would support that evaluation.

P5's suggestion of Recently Played and P6's suggestion of Library sorting are reasonable ideas to explore. They were not needed to complete the tested tasks, so they should be considered alongside further observations of how people return to content. A small prototype extension could examine whether these options help with that activity and whether their purpose is clear alongside saved and unfinished items.

6.6 Conclusion

Omnisound explored how music, podcasts and audiobooks could share search, saving and playback while keeping their content types recognisable. The project identified a focused set of interaction priorities, explored alternative layouts through early ideas and sketches, and developed two interactive prototypes. This met the design and evaluation goals and helped us understand how people used the main functions.

All participants completed the three core tasks in both usability-testing rounds. The first evaluation nevertheless showed that Continue Listening could be easier to notice and Save could be more prominent. We made focused changes to Home and saving in V2, then tested the same goals again. The second group used Continue Listening more directly and recorded shorter task times. These findings support the value of the iteration for the tested interactions, while remaining specific to the small groups and prototype scope.

The main lesson was that making a function available does not guarantee that people will find it when they need it. Watching participants use Omnisound revealed difficulties that were easy to overlook while designing the screens. Their actions and comments gave us concrete reasons to make changes and a clear focus for testing the next version.

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